2.Programming Network Communications and Services (JSON)

AIM:To study and implement Programming Network Communications and Services (JSON)

- Connect to Internet
- Fetch JSON data from server
- Parse JSON
- Display data in TextView

We will fetch JSON from:
https://jsonplaceholder.typicode.com/users

Add Internet Permission

Open **AndroidManifest.xml**

Inside <manifest> tag.

```
<uses-permission android:name="android.permission.INTERNET"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">
        <Button
            android:id="@+id/btnFetch"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Fetch JSON Data"/>
```

```
        <TextView
            android:id="@+id/tvResult"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:textSize="16sp"
            android:paddingTop="20dp"/>
    </LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.example.jsonexample;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import org.json.JSONArray;
import org.json.JSONObject;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
public class MainActivity extends AppCompatActivity {
    Button btnFetch;
    TextView tvResult;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnFetch = findViewById(R.id.btnFetch);
        tvResult = findViewById(R.id.tvResult);
```

```
        btnFetch.setOnClickListener(v -> fetchJSON());
    }
    private void fetchJSON() {
        new Thread(() -> {
            try {
                URL url = new URL("https://jsonplaceholder.typicode.com/users");
                HttpURLConnection conn = (HttpURLConnection) url.openConnection();
                conn.setRequestMethod("GET");
                BufferedReader reader = new BufferedReader(
                        new InputStreamReader(conn.getInputStream()));
                StringBuilder result = new StringBuilder();
                String line;
                while ((line = reader.readLine()) != null) {
                    result.append(line);
                }
                JSONArray jsonArray = new JSONArray(result.toString());
                StringBuilder displayData = new StringBuilder();
                for (int i = 0; i < jsonArray.length(); i++) {
                    JSONObject obj = jsonArray.getJSONObject(i);
                    displayData.append("Name: ")
                            .append(obj.getString("name")).append("\n");
                    displayData.append("Email: ")
                            .append(obj.getString("email")).append("\n\n");
                }
                runOnUiThread(() ->
                        tvResult.setText(displayData.toString()));
            } catch (Exception e) {
                e.printStackTrace();
            }
```

```
}
).start();
   }}
```


11:04 AM
92%
jsonexample
[Fetch JSON Data]
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Email: Nathan@yesenia.net
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